

Validation Agent - AI Agentic Automation Architecture

Architecture view for the AI-assisted COBOL-to-C# validation platform. This page consolidates the core component architecture, command surface, session lifecycle orchestration, and the RAG knowledge-base pattern used for automated issue analysis and fix guidance. It also presents the solution as an AI agentic automation system with autonomous remediation loops, model-driven reasoning, and governed delivery.

Presentation Layer + Orchestration

Agentic AI Control Plane

Agent and Service Layers

RAG Knowledge Base Pipeline

Session Lifecycle Management

Models: LLM, Embeddings, Prompting

Autonomous Remediation Loop

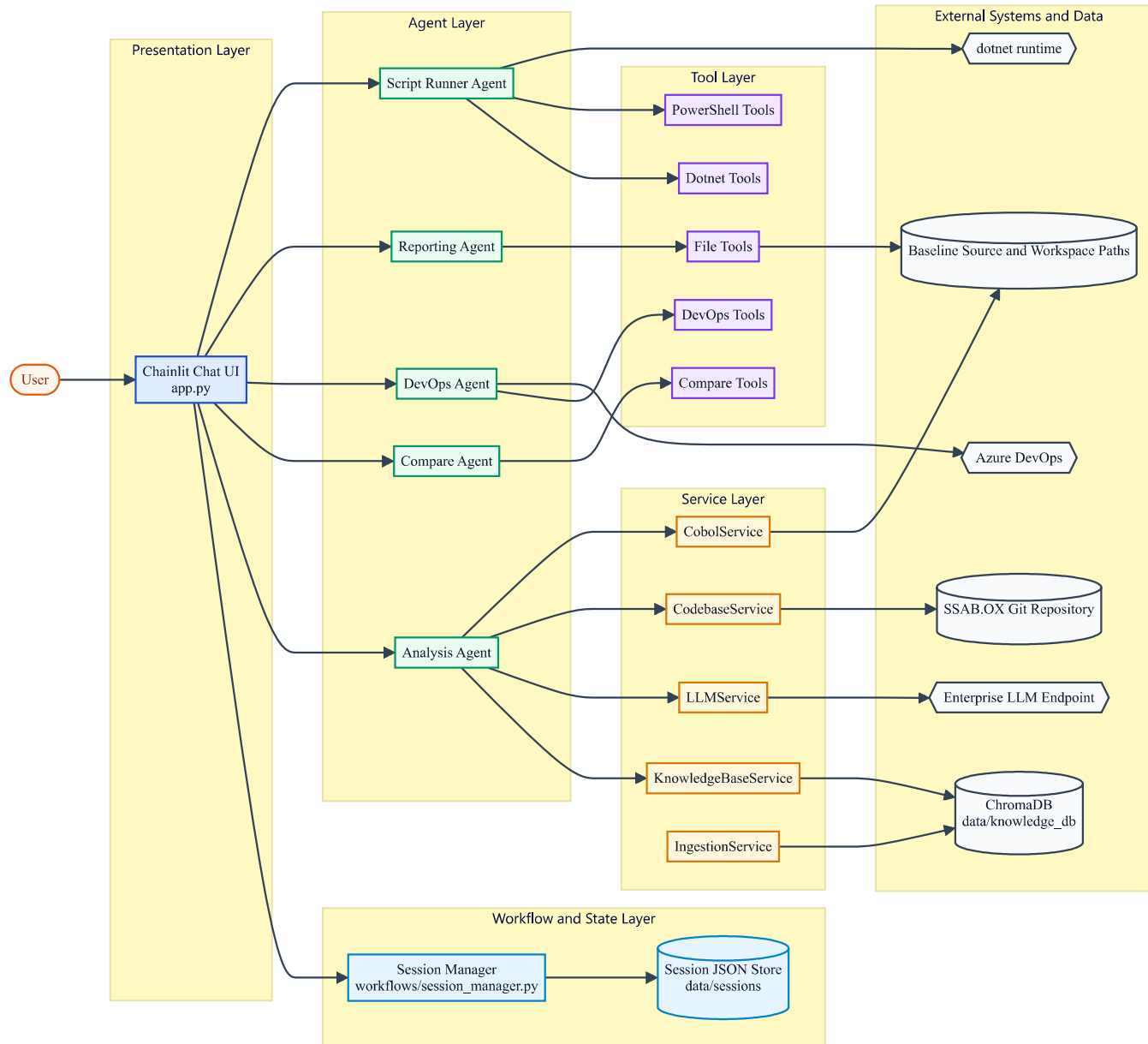
Sharing Guidance

Share this page using an HTTPS OneDrive share link. Local file URLs (file:///...) are only accessible on the same machine and cannot be opened by other users.

Recommended: Right click the HTML file in OneDrive, choose **Share**, and send the generated HTTPS link to your team. **Why:** It avoids local file restrictions and keeps Mermaid rendering consistent across users.

System Architecture Overview

High-level enterprise architecture with layered decomposition and directional flow across user interaction, agentic orchestration, autonomous agent execution, supporting services, tooling, model endpoints, and external data systems.



BLUE

Presentation and orchestration boundary.

GREEN

Specialized agents handling domain operations.

AMBER/PURPLE

Service and tooling dependencies.

Available Commands

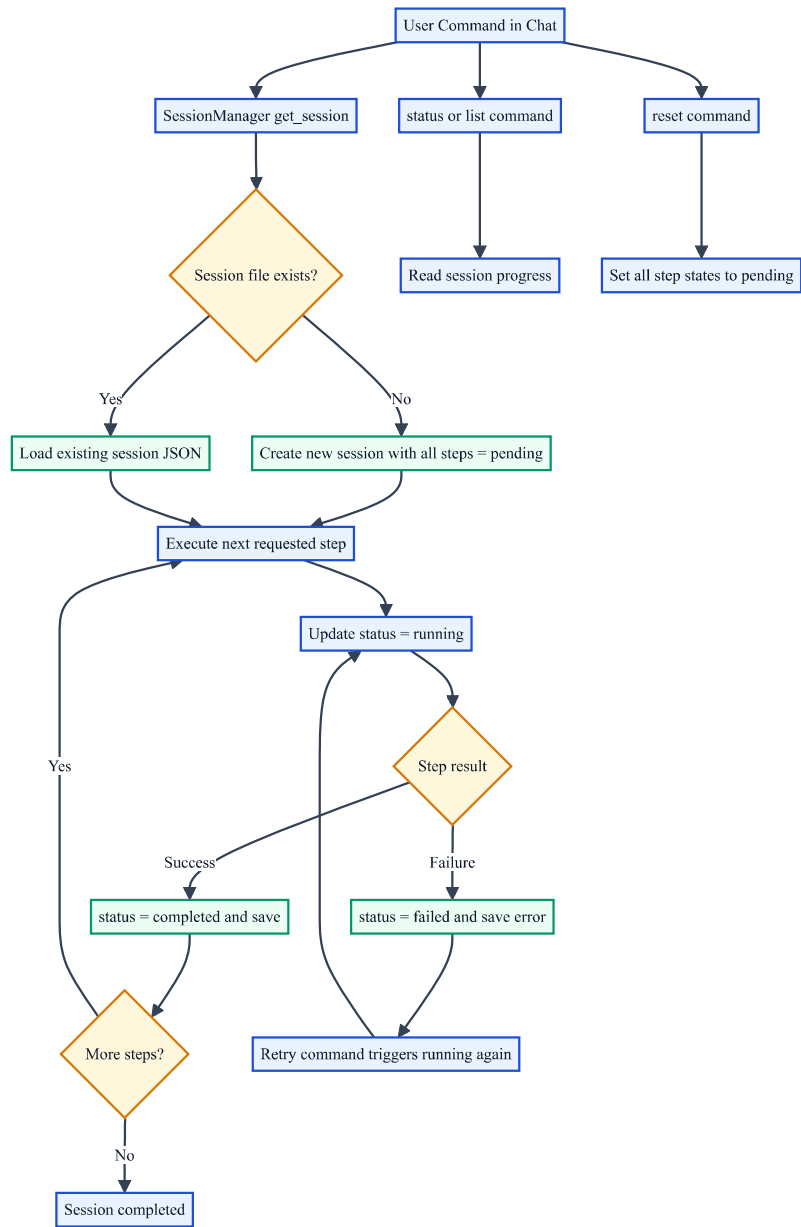
Command surface exposed in chat for human-triggered control, stage-wise orchestration, and full autonomous validation runs.

COMMAND	STEP	PURPOSE
copy <baseline>	1	Copy baseline files to workspace target path.
load <baseline>	2	Load baseline data for test execution.
branch <baseline>	3a	Create branch in source repository.
run <baseline>	3b	Run migrated C# program with dotnet runtime.
unload <baseline>	5	Unload output datasets from DB/runtime stage.
compare <baseline>	6	Compare generated outputs with legacy baseline.
report <baseline>	7	Generate report artifacts for review.
finalize <baseline>	8	Finalize and hand over for PR workflow.
validate <baseline>	All	Execute the full pipeline from copy to finalize.
reporting <baseline>	All	Run reporting-centric orchestration mode.
status <baseline>	-	Show step status of an existing session.
reset <baseline>	-	Reset a session back to pending state.
list	-	List all available baseline sessions.
help	-	Display usage and command help.

Sharing tip: Use the **Export to PDF** button for architecture reviews. **Editing tip:** Diagrams are Mermaid blocks and can be updated directly in this HTML file.

Session Management Architecture

Per-baseline checkpointed state machine with persisted sessions, resumable execution, retry progression, and agentic workflow continuity.



RAG and Knowledge Base Architecture

Detailed retrieval-augmented generation flow showing how historical fixes are indexed and how runtime failures trigger context retrieval, prompt orchestration, model reasoning, and iterative autonomous fix attempts.

